

JACOB W. RAMPINO

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EDUCATION

University of Chicago, Chicago, IL

MA, Computational Social Science; Expected June 2027

Thesis: “Detecting Antisemitism: Multi-Aspect Operationalization of a Contested Concept via Natural Language Inference”

Advisor: Dr. Isaac Mehlhaff

Florida State University, Tallahassee, FL

BS, Political Science and History with Honors; May 2025

Thesis: “Identifying and Analyzing Selectivity in News Media Coverage Through Hierarchical Topic Modeling”

Advisor: Dr. Holger Kern

RESEARCH EXPERIENCE

University of Chicago, Chicago, IL

Graduate Research Assistant, Political Science Department (January 2026 – Present)

- Design an LLM and Political DEBATE stance detection pipeline for hypothesis generation and evaluating entailment agreement across 87,000+ posts.
- Construct a randomized labeling survey on Qualtrics integrated using JavaScript to collect a ground-truth dataset for validating and fine-tuning entailment tasks.
- Clean over 9.1 million Reddit posts across 10 years by removing extraneous regular expressions, extracting features from natural language, and extracting depth and thread structure using parent-child relationships.
- Design and implement a topic modeling pipeline to cluster over 9.1 million Reddit posts, optimizing HDBSCAN parameters on a subset using a parallelized grid search on high-performance computing clusters.

Florida State University, Tallahassee, FL

Undergraduate Research Assistant, Political Science Department (August 2023 – April 2024)

- Deployed Structural Topic Models in R to analyze how topics and beliefs on immigration vary across self-identified partisan affiliation.
- Measured partisan bias across 6 major outlets by scoring article text against 2,000 Congressional speech phrases, finding up to 5.4-point ideological divergence between outlet pairs on a -6 to +6 scale.
- Generated bar charts, box plots and scatter plots to identify selection bias in source-topic trends using ggplot2.
- Hand-coded survey responses on immigration attitudes, classifying sentiment to generate labeled training data.

PROFESSIONAL EXPERIENCE

The Chicago Maroon, Chicago, IL

Data Reporter (April 2026 – Present)

- Engineered a distributed data extraction pipeline to process 83 years of historical newspaper archives, orchestrating MPI-parallelized batch jobs on the Midway3 HPC cluster to handle a 300GB+ PDF corpus.
- Designed the pipeline’s end-to-end data infrastructure, including rclone-based egress from Hetzner R2 object storage, pHash deduplication, and a SQLite database configured for concurrent write access to track per-image detection metadata such as bounding box coordinates, confidence scores, dimensions, and page identifiers across the full corpus.
- Applied the Detectron2 Newspaper Navigator model to extract figures from scanned historical print, generating CLIP embeddings and indexing approximately 1.3 million vectors with FAISS to enable multimodal image and text retrieval across the corpus.
- Scraped and compared university caffeine consumption at a national level using geographic, enrollment, and establishment data collected from IPEDS and the SerpAPI Google Maps endpoint.

SKILLS

Languages Python · R · SQL · JS · HTML · LaTeX **ML** scikit-learn · XGBoost · PyTorch **CV** CLIP · Detectron2 **NLP** sentence-transformers · BERTopic · STM · gensim · NLI **Stats** statsmodels · scipy.stats **HPC** MPI · Slurm · PySpark · rclone · joblib **Scraping** BeautifulSoup · Selenium · AT Protocol **GUI** Tkinter · Streamlit **Plots** Plotly · Matplotlib · Seaborn · ggplot2